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8-PUZZLE SEARCH PERFORMANCE REPORT

Comparative Performance Analysis of Uninformed & Informed Search Strategies

Domain: Artificial Intelligence
Problem: 8-Puzzle State-Space Search
Status: Empirical Experiment Complete

1. EXECUTIVE SUMMARY

The **8-Puzzle** is a classical sliding-block problem consisting of a 3×3 grid with 8 numbered tiles (1 through 8) and an empty space (denoted as `0`). The goal is to transform an initial tile permutation into a standard row-major target configuration by sliding tiles into the blank position.

Initial State

6	3	2
7	1	5
0	4	8

Goal State (Target)

0	1	2
3	4	5
6	7	8

This report analyzes an empirical evaluation of five search agents initialized from an identical 8-step permutation: **Breadth-First Search (BFS)**, **Depth-First Search (DFS)**, **Iterative Deepening DFS (IDFS)**, and **A* Search** with **Manhattan** and **Euclidean** distance heuristics.

2. KEY DESIGN DECISIONS & CORE ASSUMPTIONS

- **Unit Step Cost (c(s, a, s') = 1):** Every move has a uniform cost of 1. Path cost *g(n)* strictly equals search depth *d*.
- **State Immutability:** The `Puzzle8` state class is immutable; calling `move(action)` returns a fresh instance, ensuring parent references in the search tree remain unaltered.
- **Decoupled Node Metadata:** The `Node` class stores search statistics (*g*, *h*, *f = g + h*, parent pointer, incoming action) separately from domain state logic.
- **Action Perspective:** Actions describe movement of the **blank tile (0)**: `Up`, `Down`, `Left`, `Right`.
- **Canonical Goal:** Standard row-major array `[0, 1, 2, 3, 4, 5, 6, 7, 8]`.

3. INITIAL STATE GENERATION & MATHEMATICAL SOLVABILITY

3.1 Solvable Random Walk Mechanism

Random board generation uses `random_solvable_board(num_moves=8)` in `main.py`. Starting from the goal state, it executes an 8-step random walk while explicitly suppressing immediate reverse actions (e.g., `Up` followed by `Down`). This strictly guarantees that the resulting configuration is solvable within at most 8 moves.

3.2 Analytical Solvability (Inversion Parity Proof)

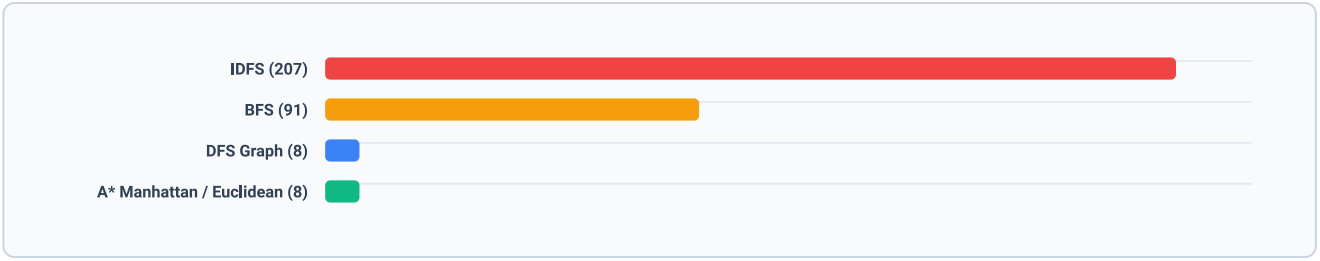
For a 3×3 grid, a state is solvable if and only if its **inversion count** is **even**. An inversion occurs when a tile *i* precedes tile *j* in row-major order such that *i > j* (ignoring blank 0).

Initial Tile Array (excluding 0): `[6, 3, 2, 7, 1, 5, 4, 8]`
• After 6: 3, 2, 1, 5, 4 → 5 inversions | • After 3: 2, 1 → 2 inversions | • After 2: 1 → 1 inversion
• After 7: 1, 5, 4 → 3 inversions | • After 1: none → 0 | • After 5: 4 → 1 | • After 4 & 8: 0
Total Inversions = 5 + 2 + 1 + 3 + 0 + 1 + 0 + 0 = 12 (12 mod 2 ≡ 0 → **Mathematically Solvable**).

4. AGENT DATA STRUCTURES & OPERATIONAL DESIGN

Agent	Frontier Data Structure	Explored Set / Cycle Prevention	Ingestion / Eviction Strategy	Theoretical Optimality
BFS	`collections.deque`	Hash Set (`seen`)	FIFO (First-In, First-Out)	Optimal (unit cost)
DFS (Graph)	Dynamic Array (`list`)	Hash Set (`explored`)	LIFO (Last-In, First-Out)	Non-Optimal
IDFS	Call Stack / `list`	Ancestor Path Check (`_on_path`)	Depth-Bounded LIFO	Optimal (unit cost)
A* (Manhattan)	Binary Min-Heap (`heapq`)	Hash Set (`explored`)	Priority: $f(n) = g(n) + h_M(n)$	Optimal (admissible)
A* (Euclidean)	Binary Min-Heap (`heapq`)	Hash Set (`explored`)	Priority: $f(n) = g(n) + h_E(n)$	Optimal (admissible)

5. PERFORMANCE VISUALIZATIONS & EXPERIMENTAL METRICS



Summary Comparison Table

Metric	BFS	DFS (Graph)	IDFS	A* (Manhattan)	A* (Euclidean)
Solution Found	True	True	True	True	True
Path Cost (g)	8	8	8	8	8
Search Depth (d)	8	8	8	8	8
Max Depth Reached	8	8	8	7	7
Nodes Expanded	**91**	**8**	**207**	**8**	**8**
Running Time (s)	0.000579 s	0.000062 s	0.001589 s	0.000104 s	0.000092 s
Guaranteed Optimal?	Yes	No	Yes	Yes	Yes

6. COMPARATIVE PERFORMANCE ANALYSIS

Informed Search & Dominance

Both A* agents expanded only **8 nodes**, following the optimal path directly. Manhattan distance ($h_M = \sum(|\Delta x| + |\Delta y|)$) is consistent and strictly dominates Euclidean distance ($h_E = \sum\sqrt{(\Delta x^2 + \Delta y^2)}$) on grid structures, as $h_E \le h_M \le h^*$.

Uninformed Search Trade-Offs

IDFS expanded 207 nodes due to iterative re-evaluations ($\sum N_i$ across depth limits 0..8), but retained O(bd) linear space. **BFS** expanded 91 nodes to reach depth 8. Graph **DFS** fortuitously expanded only 8 nodes because reversing child expansions unwound the 8-step generator.

7. TRACEABLE STEP-BY-STEP TRANSITION SEQUENCE

All five agents generated the identical optimal 8-step solution sequence: `Up → Up → Right → Down → Down → Left → Up → Up` (Cost = 8).

Initial State

6	3	2
7	1	5
0	4	8

Step 1: Move Up

6	3	2
0	1	5
7	4	8

Step 2: Move Up

0	3	2
6	1	5
7	4	8

Step 3: Move Right

3	0	2
6	1	5
7	4	8

Step 4: Move Down

3	1	2
6	0	5
7	4	8

Step 5: Move Down

3	1	2
6	4	5
7	0	8

Step 6: Move Left

3	1	2
6	4	5
0	7	8

Step 7: Move Up

3	1	2
0	4	5
6	7	8

Step 8 (Goal State Reached): Move Up

0	1	2
3	4	5
6	7	8